User Experience and Acceptance of AI

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Abstract

The use of AI technologies has proliferated across various domains, ranging from healthcare and finance to entertainment and transportation. As these technologies continue to advance, it becomes increasingly important to understand the role of user experience (UX) and acceptance in their adoption and utilization. In the study we focused specifically on AI copilot, AI voice assistant, virtual reality (VR), and augmented reality (AR). It highlights the importance of optimizing user experience and addressing acceptance factors to promote the adoption and utilization of these technologies.

In the case of AI copilot technology, user experience is crucial in facilitating effective interactions between programmers and code-generating models. The lack of transparency and explainability in the generated code poses a challenge to customization and refinement. The study emphasizes the role of user interfaces in providing insights into the behavior and reasoning of code-generating models, enabling programmers to make informed decisions and refine the code to suit their needs.

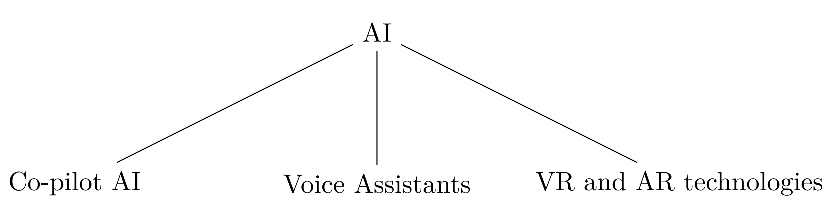
Regarding AI voice assistants, user perception and utilization are influenced by factors such as the complexity of interactions and the behavior of voice assistants. Challenges include formulating precise commands, misinterpretation of queries, and privacy concerns. Strategies to enhance user experience involve continuous learning and updates, intuitive user interfaces, personalization and contextual understanding, and robust privacy and security measures.

The abstract also mentions the impact of VR and AR on user experience and acceptance. In the case of AR, studies focus on user experience and acceptance of AR glasses for science literacy and astronaut's manual work support. The findings highlight the importance of spatial presence, simulator sickness, and perceived usefulness in determining acceptance. Improvements in UI design, field of view, and support for specific formats are suggested to enhance user experience.

Overall, prioritizing user experience, transparency, and explainability, along with addressing acceptance factors such as usefulness, ease of use, and trust, are essential for the successful integration and adoption of AI copilot, AI voice assistant, VR, and AR technologies. These advancements have the potential to improve efficiency, innovation, and societal impact across various sectors.

1 Introduction

The rapid advancement of technology has given rise to innovative solutions aimed at improving our daily lives. Among these technological developments, AI Copilot, Virtual Reality (VR) and Augmented Reality (AR), and Voice-controlled Assistants have secured considerable attention for their potential to revolutionize user experiences across various domains. AI refers to "cognitive" functions that we associate with the human mind, including problem solving and learning. Syam and Sharma 2018. Recognizing the importance of user experience and acceptance in these technologies is crucial for their successful adoption and widespread use.



AI Copilot technology, also referred to as intelligent virtual assistants or conversational agents, leverages artificial intelligence algorithms and automation to assist users in complex tasks. It is an AI-based technology that works alongside a human operator, aiding, recommendations, or taking over certain tasks to improve performance, efficiency, and safety. It is designed to complement and augment human capabilities. These intelligent systems possess the ability to comprehend user intentions, offer accurate recommendations, and provide effective assistance. By serving as virtual companions or assistants, AI Copilots have the potential to streamline workflows, reduce cognitive load, and enhance user productivity. However, optimizing user experience and addressing challenges like over-reliance on AI, lack of transparency, and managing exceptional cases are essential for the successful implementation of AI Copilot technology.

VR and AR technologies deliver immersive and interactive experiences by blending virtual elements with the real world or simulating entirely virtual environments. These technologies have found applications in diverse fields, including gaming, education, architecture, healthcare, and training. User experience considerations in VR and AR encompass factors such as high-quality visuals, intuitive interaction design, motion sickness mitigation, and fostering an overall sense of presence. Well-designed VR or AR experiences have the power to transport users to virtual worlds, offer realistic simulations, and facilitate enhanced learning and engagement. However, addressing challenges related to hardware limitations, content creation, and user discomfort is crucial for ensuring optimal user experiences.

Voice-controlled Assistants have gained popularity through the increasing utilization of natural language processing and voice recognition technologies. These assistants enable users to interact with digital systems through voice commands, providing hands-free operation and efficient access to information and services. User experience considerations for voice-controlled assistants involve ensuring accurate speech recognition, natural language understanding, and system responsiveness. Seamlessly integrating these assistants with other devices and services enhances user experience, delivering convenience and personalized interactions. Nevertheless, challenges related to speech recognition accuracy, privacy concerns, and limited functionality in specific domains can impact the overall user experience.

To foster the successful adoption and integration of AI Copilot, VR and AR, and Voice-controlled Assistant technologies, prioritizing user experience and acceptance is paramount. Designing intuitive interfaces, optimizing system performance, and addressing user concerns regarding privacy, security, and trust are crucial elements in enhancing user experiences. Additionally, transparent design practices, clear communication of data usage, and providing sufficient support and training are instrumental in promoting user acceptance and driving widespread adoption of these technologies.

This report delivers the overall user experience and acceptance within the context of AI Copilot, VR and AR, and Voice-controlled Assistant technologies. By exploring the unique considerations and challenges associated with each technology, our aim is to shed light on the significance of user experience and acceptance for their successful implementation and broader usage. Through an analysis of existing research findings, case studies, and insights from industry experts, we identify key factors influencing user experience and acceptance, while offering recommendations for optimizing the design and deployment of these technologies.

2 User Experience and Acceptance

2.1 AI Copilot Technology

Large language models (LLMs), such as GPT-3, have enabled the development of AI-powered code assistants like Copilot. These code recommendation systems extend existing IDE code completion mechanisms and utilize neural models trained on billions of lines of codeMcNutt et al. 2023; Vaithilingam, Zhang, and Glassman 2022. The goal of building this type of AI assistant is to accelerate the development process by reducing workloads Mozannar et al. 2023 but the strength and weaknesses of these tools are not analyzed properly preventing them from being used optimally Döderlein et al. 2023.

2.1.1 User Interaction with Code Recommendation Systems

- Programmer Behavior by Task and Expertise
  Programmers interact with code recommendation systems by utilizing various programming states, including acceleration and exploration modes Barke, James, and Polikarpova 2022. In the acceleration

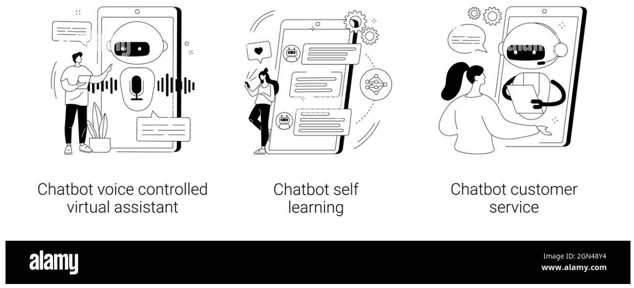


Figure 1: Copilot chatbot

mode, programmers quickly achieve their intended tasks with Copilot's assistance, while the exploration mode involves more deliberate interaction, explicit prompting, and extensive validation Barke, James, and Polikarpova 2022; Ross et al. 2023.

- User Interaction Patterns with Copilot
  User interaction with Copilot involves generating multiple lines of code, which can cause cognitive overload, especially when the generated code contains errors. Constant context switching between reading, writing, and debugging creates significant mental pressure for programmers Mozannar et al. 2023.

2.1.2 User Perception and Utilization of Copilot

Despite Copilot not necessarily improving task completion time or success rate, the majority of participants expressed a preference for using Copilot in their daily programming tasks. Copilot was found to be a useful starting point, saving users the effort of searching for solutions online Mozannar et al. 2023; Vaithilingam, Zhang, and Glassman 2022.

2.1.3 Challenges and Obstacles

- Difficulties in Understanding and Debugging Generated Code
  Novice programmers face difficulty to plan the composition during the interaction with copilot Jayagopal, Lubin, and Chasins 2022. Participants faced challenges in understanding and working with long blocks of code generated by Copilot. This necessitates better approaches for code editing, debugging, and code repair Barke, James, and Polikarpova 2022.

- Ambiguity in Using Comments as Specifications
  Using comments as a means to communicate with Copilot can be ambiguous, as slight changes in comments can significantly alter the generated code snippets. This poses a challenge in providing precise instructions to Copilot

- Overestimating Copilot's Code Quality and Debugging Effort
  Participants often underestimated the effort required to fix bugs in the code generated by Copilot. They encountered difficulties in debugging and verifying the correctness of the generated code, leading to additional time spent on code repair. Since AI models for code generation are trained on large public repositories, there is a possibility that low-quality training data can influence models to suggest low-quality code or code with security vulnerabilities Sarkar et al. 2022.

2.1.4 Improving the Design and User Experience of Copilot

Based on user feedback and observations, several potential directions for improving Copilot's design were identified. These improvements include suggesting short and high-confidence code when the user is in acceleration mode Barke, James, and Polikarpova 2022, better approaches for code editing, debugging

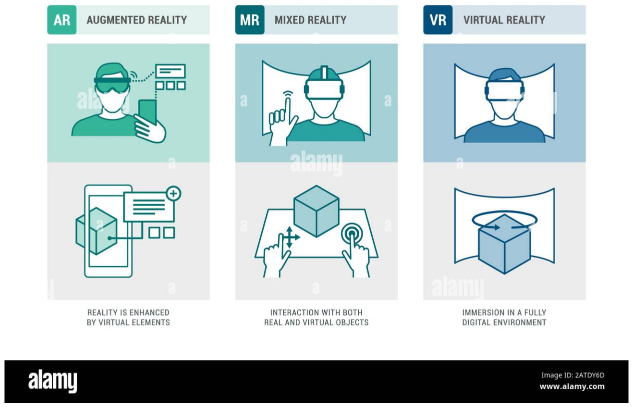


Figure 2: VR and AR Technology

support, explanations for generated code, and automatically generating test cases and data for code validation .

2.2 Augmented Reality(AR) and Virtual Reality(VR)

Augmented and virtual reality (AR & VR) seems to have a lot of promise for training and education. Consumer-grade augmented and virtual reality glasses are starting to become available, despite the fact that this technology has already gained popularity thanks to handheld mobile devices. By enhancing mobility, performance, and safety, these glasses could unleash the full potential of augmented and virtual reality.

2.2.1 User Experience and Technology Acceptance in Augmented Reality Science Literacy

A recent report article addressed how users will react to learning about the new technology conducting experiments using augmented reality technology on a population of university students. Vrellis et al. 2020 This study explores user experience and acceptance of state-of-the-art AR glasses. A simple activity was created to visualize electromagnetic radiation for science literacy purposes and then evaluated by university students. For this study the Magic Leap One AR headset was used. This device is a combination of augmented reality see-through glasses and a standalone computer. It runs on Lumin OS which is a spatial operating system that can recognize and understand different environments, power high-fidelity visuals, and ensure that digital content mixes appropriately with the real world. The device utilizes state-of-the-art hardware such as a 64-bit CPU, 256 CUDA cores GPU, 128 GB storage, and a number of self-contained, head-mounted sensors and cameras. It is a standalone device that doesn't require a connection to a PC. The activity was created to visualize electromagnetic fields emitted by various everyday devices (cell phone, DECT phone, laptop, and router). The electromagnetic waves were visualized as a set of concentric spheres centered on each device and expanding in all directions. The devices were placed in specific locations and the virtual waves were rendered on the corresponding 3D coordinates using the space mapping capabilities of the headset. A total of 154 undergraduate students of the Department of Primary Education, University of Ioannina, Greece participated in this study. Their ages ranged from 19 to 47 years (Mean = 21.94, SD = 4.72), and most of them (83.1Although participants felt that the virtual electromagnetic waves were blended in the real environment, Spatial Presence was found to be moderate, maybe because of the simplicity of the digital content and the lack of interactivity. Simulator sickness was generally low, indicating that the AR glasses were well tolerated by the participants as found in a similar study. No significant correlation was found between Spatial Presence and Simulator Sickness. According to TAM, the participants considered the use of AR as a good idea for visualizing abstract physics concepts. They thought that it was useful and easy and increased their awareness of their physical environment. Consequently, their attitude towards this novel technology was very positive and they would like to use it in the future. Acceptance seemed to be positively correlated with Spatial Presence and negatively correlated with Simulator Sickness.

2.2.2 Augmented Reality System for Astronaut's Manual Work Support

Another article on the use of augmented reality technology for astronaut's manual task support puts light on the question of astronauts' user experiences with becoming acquainted with this new technology. Helin et al. 2018 Augmented Reality (AR) system to support an astronaut's manual work, has been developed in two phases. The first phase was developed in the Europeans Space Agency's (ESA) project called "EdcAR—Augmented Reality for Assembly, Integration, Testing and Verification, and Operations" and the second phase was developed and evaluated within the Horizon 2020 project "WEKIT—Wearable Experience for Knowledge Intensive Training." main aim is to create an AR-based technological platform for high knowledge manual work support, in the aerospace industry with reasonable user experience. The AR system was designed for the Microsoft HoloLens mixed reality platform, and it was implemented based on a modular architecture. Smart Glasses User Satisfaction questionnaire was created for the WEKIT trials. SGUS measures subjective satisfaction focusing especially on test participants' experiences of the features that support learning. SGUS is based on the evaluation criteria for web-based learning by Ssemugabi and de Villiers (2007) and statements taken from Olsson (2013) "Concepts and Subjective Measures for Evaluating User Experience of Mobile Augmented Reality Services." Based on the observation and expert evaluation HoloLens-based AR-system is usable for basic daily use as its usability has reached a reasonable level. Helin et al. 2018. The 3D space user interface and annotations were working properly without any delay and the image quality was high enough for see-through glasses. The main downside of the system is the narrow field of view, the user has to move his/her head around and search for the information in some cases. Test subjects also highlight the potential of applying AR technology to this field, but AR-systems UI should be updated to support Operations Data File (ODF) format which is common in the space domain.

2.2.3 User Experience of AR & VR in retail industry

The integration of virtual reality (VR) and augmented reality (AR) technologies is transforming the retail industry by offering enhanced customer experiences. These technologies enable online product customization, interactive visualization, and immersive shopping journeys, capturing customer attention and influencing purchasing decisions. Retailers, including Alibaba, LensKart, Nykaa, and Amazon, have implemented dedicated VR/AR platforms to provide interactive and immersive shopping experiences. Kumar 2021. Through these platforms, customers can select products based on personalized virtual visualizations, leading to increased engagement and brand loyalty. Additionally, VR and AR technologies have found applications beyond retail in sectors such as tourism and museums, revolutionizing user experiences and influencing customer buying behavior. However, challenges such as cost, privacy protection, and technical complexities need to be addressed to ensure widespread adoption and successful implementation of VR and AR technologies. The integration of VR and AR technologies represents a significant opportunity for retailers to create captivating and immersive shopping experiences. By leveraging these technologies, retailers can enhance brand image, engage customers, and influence their purchasing decisions. Kumar 2021. The ability to offer personalized product visualizations and seamless interactions between customers and virtual items provides a competitive edge in the retail industry. Moreover, the impact of VR and AR extends beyond retail, shaping user experiences in various sectors. While challenges such as cost and privacy concerns must be addressed, the potential for VR and AR technologies to transform the retail industry and enhance customer engagement is immense. As technology continues to evolve, retailers need to embrace these innovations to stay ahead in an increasingly competitive market and provide exceptional shopping experiences to their customers. Overall Improving the user experience of VR & AR will give advantage in retail industry in lot of ways like Improved Engagement, Enhanced Product Understanding, Increased Convenience, Emotional Connection, Evolving Customer Expectations

2.3 Voice assistants Technologies

Voice assistants are software applications that leverage artificial intelligence (AI) to understand and respond to user commands and queries through voice interaction. They utilize natural language processing (NLP), machine learning, and voice recognition technologies to interpret and execute user requests. Some popular examples of voice assistants include:

1. Alexa by Amazon

Figure 3: Voice Assistant

2. Siri by Apple
3. Google Assistant by Google

2.3.1 Using Complexity of Voice Assistants

Voice assistant interactions can be complex, primarily due to variations in language, accents, and user preferences. This complexity may result in challenges for users, such as difficulty in formulating precise commands or the voice assistant misunderstanding user intents. Non-native speakers informed that they wanted the VA's to read out the texts at a more slower pace, and in a more natural way. For task number 4 many of the non-native speakers reported a problem in making the VA's understand some music or artist name that are non-English. Pal et al. 2019 . These challenges can lead to frustration and hinder user acceptance of voice assistants.

2.3.2 Behavior of Voice Assistants

The behavior of voice assistants plays a crucial role in determining user experience and acceptance. Users expect accurate and relevant responses to their commands and queries. Voice assistants employ NLP algorithms to understand user inputs, analyze context, and generate appropriate responses. Improving the behavior of voice assistants involves refining language understanding capabilities, training models with extensive datasets, and continuously updating algorithms to enhance accuracy and effectiveness. For example, Users found the praise from a male-voiced computer more compelling than the same comments from a female-voiced computer Nass and Moon 2000

2.3.3 Problems with Voice Assistants

Despite their advancements, voice assistants still face certain problems that impact user experience and acceptance. Some common issues include misinterpretation of commands, especially in noisy environments or with complex queries or finding names of which are not in English language. Pal et al. 2019 . Privacy concerns regarding the storage and handling of user data also pose challenges. Even with stricter vetting processes for Alexa kid skills, children users remain susceptible to risks such as private data collection and exposure to unsuitable content. Le et al. 2022. Approximately 43% of participants did not believe that Alexa could leak private information, and 72% of participants did not think it could have a negative influence on their children. Yang et al. 2023 . Furthermore, limited functionality and inability to understand specific commands or context can frustrate users and impede acceptance.

2.3.4 Ways to Solve Problems and Enhance User Experience

To address the problems associated with voice assistants and improve user experience and acceptance, several strategies can be implemented:

- Continuous Learning and Updates:
  Voice assistants should undergo regular updates to improve language understanding and expand their knowledge base. Machine learning techniques can be employed to analyze user feedback and refine algorithms accordingly.

- Intuitive User Interfaces and Interaction Models:
  Designing user interfaces that facilitate seamless and intuitive interactions with voice assistants is crucial. Clear instructions, prompts, and feedback can guide users in formulating precise commands and reduce frustration

- Personalization and Contextual Understanding:
  Enhancing voice assistants' ability to personalize responses based on user preferences and context can significantly improve user satisfaction. Contextual understanding enables voice assistants to provide more accurate and relevant information. Experience can be enhanced by, user selection of different voices and personalities for different tasks, moods, children, guests, times of the day, rooms, or device types. Zargham et al. 2021

- Robust Privacy and Security Measures:
  Addressing privacy concerns is essential for gaining user trust. Implementing stringent privacy and security measures, such as end-to-end encryption, transparent data handling practices, and user consent mechanisms, can help mitigate these concerns. Yang et al. 2023

2.3.5 Future Trends and Developments

Voice assistants are constructed based on the main speech user interface heuristics and themes of information quality and relevance, semantics and intelligence and user satisfaction. Zwakman et al. 2020 . The field of voice assistants is rapidly evolving, and several advancements are expected to shape their future. Natural language understanding (NLU) techniques, sentiment analysis, and emotion recognition are areas of active research. These developments hold the potential to significantly enhance voice assistants' ability to understand and respond to user commands, leading to more personalized and emotionally intelligent interactions.
To ensure they were capable of answering personality related questions, participants of a survey were instructed to ask the VA social and emotional related questions, such as "How is the weather today?", "How old are you?", "What is your gender?", "Are you jealous?", "Find some information about Apple Siri," "Send a text message to X," "Are you smart?", and "Play a song from Lady Gaga or Sia." Participants interacted with one of the VA mobile applications for 7 min, then closed the mobile application, and answered the survey's questions. The survey's questions explored personality traits, perceived control, consumers' focused attention during voice interaction with VA, consumers' exploratory behavior, consumers' satisfaction, and consumers' willingness to continue using VA. Poushneh 2021 These kind of anticipated improvements in user experience and acceptance will likely drive the widespread adoption of voice assistants.

3 Conclusion

In conclusion, the user experience and acceptance of AI Copilot, AI voice assistants, and AR & VR technology play a crucial role in their successful adoption and utilization. Regarding AI Copilot, despite not significantly improving task completion time or success rate, programmers expressed a preference for using it due to its usefulness as a starting point and its ability to save them the effort of online searches. However, challenges related to understanding, editing, and debugging generated code need to be addressed to enhance the overall user experience. Improvements in code editing, debugging support, explanations for generated code, and automated test case generation were identified as potential directions for enhancing Copilot's design.

For AI voice assistants like Siri, Alexa, and Google Assistant, the complexity of interactions, including variations in language, accents, and user preferences, can pose challenges for users, impacting their experience and acceptance. This can result in difficulty in formulating precise commands and lead to frustration. To improve user experience and acceptance, it is crucial to refine the behavior of voice assistants by enhancing language understanding capabilities, continuously updating algorithms, and addressing privacy concerns.

Strategies such as continuous learning and updates, intuitive user interfaces, personalization, and robust privacy measures are essential for enhancing voice assistants.

Similarly, in the case of AR & VR technology, user experience is a key factor in its adoption and acceptance. Evaluations in the context of science literacy showed that participants had a positive attitude toward AR technology, considering it useful and easy to use. While the blending of virtual electromagnetic waves with the real environment was perceived well, limitations such as a narrow field of view and the need to search for information were highlighted. Updating the user interface to support specific file formats was suggested to improve usability in the aerospace industry.

In summary, improving the user experience and addressing specific challenges are crucial for the acceptance and effective utilization of AI Copilot, AI voice assistants, and AR & VR technology. By focusing on enhancing user satisfaction, addressing usability issues, and implementing user-centric design improvements, these technologies can overcome obstacles, gain wider acceptance, and deliver enhanced experiences to their users.

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